

[ENTER COMPANY NAME]

QUALITY MANAGEMENT SYSTEM

STANDARD OPERATING PROCEDURE

Risk Management Program

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Applicable Regulatory Standards	ISO 14971:2019 (Complete Lifecycle) ISO 13485:2016 Clause 7.1 EU MDR 2017/745 Annex I (GSPR)

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1.0 Purpose

This Standard Operating Procedure (SOP) defines the mandatory structural framework for identifying, assessing, controlling, monitoring, and auditing risks associated with medical devices throughout their complete lifecycle—from initial conceptual design to market commercialization, field distribution, and final decommissioning. The ultimate goal is to ensure that all potential biological, toxicological, physical, operational, and clinical hazards are systematically mitigated to an acceptable level, balancing residual risks against verified clinical benefits.

2.0 Scope

This procedure is universally applicable to all design configurations, platform modifications, clinical accessories, packaging profiles, and manufacturing processes within the corporate portfolio. The requirements map directly across pre-market research and development phases (Design Controls) as well as post-market product lifecycles (Surveillance and Vigilance loops).

3.0 Responsibility and Authority

3.1 Risk Management Team (RMT)

A multi-disciplinary committee explicitly assembled for each product family, consisting of design engineers, material specialists, clinical affairs professionals, manufacturing experts, and quality technicians. The RMT holds collaborative authority to populate and approve product risk matrices.

3.2 Lead Design/Risk Engineer

Responsible for identifying structural hazards, calculating mathematical evaluation scores, mapping prospective hazard sequences, and drafting the formal product Risk Management File (RMF).

3.3 Medical/Clinical Expert

Responsible for verifying the severity scoring metrics assigned to clinical hazards, assessing user safety parameters under real-world clinical usage, and drafting the clinical justification regarding the final Risk-Benefit Analysis (RBA).

4.0 Procedure

4.1 Risk Management Planning

At the start of any new design project, the RMT must create a dedicated product Risk Management Plan. This plan must formally specify the precise scope of the device, establish the baseline parameters for acceptable risk thresholds based on state-of-the-art benchmarks, specify the multi-disciplinary team assignments, and define the lifecycle tracking milestones.

4.2 Quantitative Risk Evaluation Framework

Risks must be evaluated using a standardized numerical grid calculated via two primary criteria: **Probability of Occurrence (P)** and **Severity of Harm (S)**. Both dimensions utilize a standardized 1 to 5 linear scale:

Evaluation Level	Probability of Occurrence (P) Criteria	Severity of Harm (S) Clinical Impact
1	Incredible / Rare: Less than 1 in 100,000 units distributed.	Negligible: Minor discomfort, no functional tissue damage, simple self-limiting events.
2	Remote: Occurs occasionally in similar device platform histories.	Minor: Reversible injury, temporary localized trauma, minor clinical treatment needed.
3	Occasional: Individual lot variability might cause sporadic instances.	Serious: Significant impairment, tissue injury requiring clinical or surgical intervention to fix.
4	Probable: Expected to occur spottily during normal field operation cycles.	Critical: Permanent structural loss, life-altering trauma, severe operational impairment.
5	Frequent: High likelihood of occurrence without system controls.	Catastrophic: Patient or user death, or complete irreversible system failure.

4.3 Risk Control Priority Order Matrix

When an identified hazard score falls into an unacceptable risk region, the RMT must systematically apply risk mitigation controls. The options must follow a strict statutory hierarchy of priority:

Priority 1: Inherent Safety by Design

Eliminate the hazard completely by modifying the physical design blueprint, changing component geometries, or shifting to inherently safer, biocompatible raw materials.

Priority 2: Protective Measures & Safeguards

Incorporate active or passive safety infrastructure, such as isolation circuits, physical shields, mechanical pressure relief valves, or automated software error traps.

Priority 3: Information for Safety

Detail clear warnings, explicit user limits, contraindications, and mandatory training requirements within the product Instructions for Use (IFU) and on the external physical labels.

REGULATORY MANDATE (ISO 14971): Risk reduction through instructions, training, or labeling (Priority 3) is considered the weakest form of mitigation. Auditors require documented engineering rationale proving that design-based options (Priority 1 or 2) were technically unfeasible before accepting labeling as the primary control mechanism.

4.4 FMEA and Risk Assessment Documents

The RMT must maintain dynamic Risk Assessment matrices, specifically including Design Failure Mode and Effects Analyses (DFMEA) and Process Failure Mode and Effects Analyses (PFMEA). All mitigation controls must be physically implemented into active technical prints and manufacturing routing steps. Following implementation, the RMT executes a post-mitigation evaluation to calculate the final residual risk level.

4.5 Risk-Benefit Analysis (RBA) and Market Review

If specific individual risks remain within a high or borderline region despite applying all technically feasible controls, the device cannot be released until a formal Risk-Benefit Analysis is executed. Clinical experts must compile peer-reviewed literature and clinical data to demonstrate that the medical benefits provided by the device's therapeutic or diagnostic outcomes significantly outweigh any residual safety risks. The final output of the pre-market risk loop is documented in a formal Risk Management Report, which must be signed and authorized prior to commercial product release.

5.0 Post-Market Risk Surveillance Loops

The Risk Management File is a dynamic, living document. On a defined annual or semi-annual review cycle, the RMT must evaluate real-world feedback data from the Post-Market Surveillance and Complaint Intake modules. If a field event reveals a completely new hazard mode, or shows that a known risk occurs more frequently or with greater severity than originally estimated, the RMF must be updated, and the affected design profiles must be re-evaluated.

6.0 Associated Reference Materials

- **SOP-QA-002:** Change Control System
- **SOP-RD-005:** Design Controls and Technical File Architecture
- **SOP-RA-004:** Post-Market Surveillance (PMS) Program
- **Form-QA-007A:** Product Risk Management Plan & File Template

7.0 Revision History Log

Revision Level	Effective Date	Author / Originator	Detailed Description of Modifications
00	[Date]	[Name], Lead Risk Engineer	Initial baseline release of formalized Risk Management Program Standard Operating Procedure aligned with ISO 14971:2019.